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ORIGINAL ARTICLE

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Validation of InBody 770 bioelectrical impedance analysis compared to a four-compartment model criterion in young adults

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Abstract

Background: Multi-frequency bioelectrical impedance analysis (MF-BIA) offers enhanced body composition outcomes in a time-efficient manner. The accuracy of stand-up MF-BIA compared against a four-compartment (4C) criterion lacks evidence.
Objectives: To validate a stand-up MF-BIA compared to a 4C criterion for fat mass (FM), fat-free mass (FFM) and body fat percentage (%fat).
Subjects/Methods: Eighty-two healthy (32% men) normal-weight (BMI: 18.5–24.9 kg/m^2) young adults were measured for body composition determined from a stand-up MF-BIA and 4C model. Validity statistics included total error (TE) and standard error of the estimate (SEE) to examine prediction error between methods.
Results: For the total sample, prediction error was the highest for %fat (TE = 4.2%; SEE = 3.9%) followed by FM (TE = 2.4 kg; SEE = 2.2 kg) and FFM (TE = 2.4 kg; SEE = 2.2 kg). In men, %fat (TE = 2.5%; SEE = 2.2%) and FM (TE = 1.9 kg; SEE = 1.6 kg) were ideal; FFM was similar to FM (TE = 1.9 kg; SEE = 1.6 kg). In women, %fat (TE = 4.7%; SEE = 4.4%) ranged from good to fairly good, and FM was very good to excellent (TE = 2.6 kg; SEE = 2.4 kg); FFM was similar to FM (TE = 2.6 kg; SEE = 2.3 kg).
Conclusions: Stand-up MF-BIA may overestimate %fat and FM, and underestimate FFM compared to a 4C model. FM and FFM estimates from MF-BIA demonstrate good agreement to a 4C model and may be a practical measure of body composition in normal-weight adults. The highest error was seen in %fat for both sexes, with greater error in women.

KEYWORDS
body composition, fat mass, fat-free mass, per cent body fat, sex differences

1 | INTRODUCTION

Body composition technology is rapidly advancing with new methods being implemented for research and clinical care. A four-compartment (4C) model is widely accepted as the gold standard for evaluating body composition (Heymsfield & Waki, 1991). By measuring various compartments of fat-free mass (FFM), including total body water (TBW) and bone mineral content (BMC), the 4C model produces the most valid estimates of total body composition (Wang et al., 2002). However, assessment of body composition with a 4C model is costly, time intensive, and may not be widely accessible. Advancements in testing methodology have increased in the

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utility for assessments to be completed quickly, while posing minimal burden and risk to subjects.

Two-compartment (2C) methods are more commonly used to assess body composition due to the ease of use and short time demand. Some 2C models, such as bioelectrical impedance analysis (BIA), also produce segmental estimates of fat mass (FM) and FFM, which may be advantageous for tracking changes over time. Semi-portable multi-frequency tetra polar bioelectrical impedance analysis (MF-BIA) devices may be a cost-effective and easy-to-use method when higher standard criterion devices are not available (Kyle et al., 2004); however, the accuracy of these new technologies is largely underexplored, particularly in comparison to multi-compartment gold standard methods. While MF-BIA devices may be feasible alternatives to extensive 4C body composition measurement, previous studies have shown their accuracy may vary based on sex, leanness, ethnicity, hydration status and physical activity (Gibson et al., 2008; Lukaski, 1997; Modlesky et al., 1996; Tinsley et al., 2020; Tinsley et al., 2017). To improve the generalizability and utility of 2C models, validation of these technologies against a multi-compartment criterion is warranted.

Current research validating MF-BIA technology against a 4C criterion method is limited (Gibson et al., 2008; Graybeal et al., 2020). To date, we are not aware of a comparison of this MF-BIA device (InBody 770; Frequencies: 1, 5, 250, 500 and 1000 kHz) to a multi-compartment criterion in a healthy, college-aged population. Data have compared similar MF-BIA devices (Seca mBCA; full range frequencies 1–500 kHz) to a 4C model in male and female body builders, reporting MF-BIA (Seca) is not a valid method to assess total body per cent fat (%fat), FM and FFM estimates (Graybeal et al., 2020). Two previous studies evaluating the validity of a MF-BIA devices against DXA in male and female athletes also demonstrated poor validity for body composition outcomes (Brewer et al., 2019; Esco et al., 2015); higher error reported in these previous studies may be related to higher muscle mass and lower adiposity in these populations (Forslund, 1998). Total body water accounts for 50–75% of body weight and is directly influenced by the amount of muscle mass (Ritz et al., 2008); higher muscle mass alters standard hydration assumptions in BIA techniques. In single-frequency BIA devices, the literature suggests the error associated with body composition measurement is beyond acceptable (Andreoli et al., 2004; Graybeal et al., 2020; Van Marken Lichtenbelt et al., 2004). It was hypothesized that this device, which employs multiple frequencies, may result in better body composition estimates and provide more valid results, in comparison to similar bioelectrical devices with a single frequency (Alves et al., 2014). With improved technology and portability, as well as the use of a wider range of frequencies, validation of this MF-BIA device against a multi-compartment criterion in a larger sample size of normal-weight individuals is warranted. The purpose of this study was to assess the validity of total body measures (%fat, FM and FFM) estimates from a semi-portable stand-up MF-BIA device against a 4C criterion in healthy young men and women.

2 | METHODS

2.1 | Subjects

Eighty-two healthy men (n = 26) and women (n = 56) ages 18–22 years old who were normal weight (BMI: 22.2 ± 1.8 kg/m^2; n = 66 Caucasian, n = 11 Asian, n = 3 Hispanic, n = 2 African American) participated in a parent study (Table 1). Data presented from these subjects were planned as a secondary outcome of the larger study. For inclusion, subjects were required to be a student between the ages of 18–22 years old with a normal-weight body mass index (BMI: 18.5–24.9 kg/m^2), had no prior history of cardiovascular, metabolic, or neuromuscular disease, and had not lost or gained 3.64 kg or more over the previous two months. All subjects were enrolled as undergraduates at a University in the United States, and data were collected during the fall semester (2018). All procedures used in the study were approved by the University's Biomedical Institutional Review Board, and all participants provided written consent upon arrival to the laboratory.

2.2 | Experimental Design

Subjects reported to the laboratory for a single testing session; measurements were completed in the same order. Subjects were required to arrive to the laboratory a minimum of eight hours fasted from food, caloric beverages, caffeine, alcohol and tobacco, but remain euhydrated. Additionally, subjects were required to abstain from strenuous exercise a minimum of twelve hours prior to testing. Inclusion criteria and adherence to pre-assessment guidelines were confirmed at the beginning of the visit. Height (Stadiometer: Perspective Enterprises, Portage, MI) and weight from a calibrated scale (Tanita Inc.,) were collected upon arrival to the laboratory; subjects wore light, athletic clothing. Body composition was determined using a four-compartment (4C) model involving compartmental measures taken from air displacement plethysmography (ADP), bioelectrical impedance spectroscopy (BIS) and dual-energy x-ray absorptiometry scan (DXA), and stand-up multi-frequency bioelectrical impedance analysis (MF-BIA) was also used to assess body

TABLE 1 Subject anthropometric characteristics [mean ± standard deviation (SD)]

	Age (years)	Height (cm)	Weight (kg)	BMI (kg/m^2)
Total Sample (n = 82)	19.6 ± 1.2	168.5 ± 9.3	63.0 ± 8.6	22.1 ± 1.8
Men (n = 26)	19.6 ± 1.1	177.8 ± 8.3	70.2 ± 8.6	22.1 ± 1.9
Women (n = 56)	19.6 ± 1.2	164.1 ± 5.9	59.7 ± 6.3	22.1 ± 1.8

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composition. For all measurements, subjects were free from metal (i.e. zippers, jewellery, hard plastic) to avoid interference with data collection accuracy.

2.3 | Procedures

2.3.1 | Body Composition

A four-compartment model criterion (4C) described by Wang et al., (2002) (Equations 1-3) was utilized to determine total body fat mass (FM), per cent body fat (%fat) and fat-free mass (FFM):

$$FM\,(kg) = 2.748\,(BV) - 0.699\,(TBW) + 1.129\,(Mo) - 2.051\,(BM) \quad (1)$$

$$\%\,fat = (FM/BM) * 100 \quad (2)$$

$$FFM\,(kg) = BM - FM \quad (3)$$

where body volume (BV) and body mass (BM) were assessed through ADP and a calibrated scale, total body water (TBW) from BIS, and total body bone mineral (Mo) was calculated using total body bone mineral content (BMC) from DXA (Equation 4).

$$Mo = BMC^{*}1.0436 \quad (4)$$

A full-body DXA scan (Lunar iDXA; General Electric Medical Systems Ultrasound & Primary Care Diagnostics, enCORE Software Version 16, General Electric, Madison, WI, USA) was used to determine BMC (kg) in order to determine Mo. Subjects were positioned on their back in the centre of the scanning table laying supine with their palms facing inward. Feet were set hip-width distance apart; a Velcro strap was used to ensure that all subjects had the same distance between their legs. Height, weight, sex, ethnicity and date of birth were entered into the DXA software. All scans and region of interest analysis were conducted by a certified DXA research team member.

Total body water was estimated from BIS (SFB7; ImpediMed) for the 4C criterion (Buendia et al., 2015; Moon et al., 2008). Subjects were asked to lie supine for five min prior to testing and remove their right footwear. The subject's right wrist and right foot were cleaned with rubbing alcohol; two electrodes were then placed on the right wrist (between the distal end of the radius and ulna, and five centimetres away from this electrode proximal to the third metacarpal-phalangeal joint) and right foot (between the malleoli of the ankle, and five centimetres away from this electrode proximal to the third metatarsal–phalangeal joint). Height, weight, sex and age were entered into the device. The average of two measurements was considered for TBW.

Body volume was estimated from ADP (BodPod; Cosmed, Software Version 5.4.1, Concord). Subjects wore spandex or swimsuit-like clothing, and swim cap was worn over their head to minimize the effect of isothermal air. Subjects were asked to sit upright inside the testing chamber and breathe normally. Standard prediction equations were utilized from the software to predict thoracic gas volume (Collins & McCarthy, 2003; Demerath et al., 2002). Body volume was measured by a minimum of two trials that were within 150 mL of each other.

MF-BIA (InBody 770; Biospace Co.,) was also used to estimate total body composition: %fat, FM, and FFM. After standing for 5 min (within manufacturer guidance), subjects stood barefoot on the device's scale with the soles of their feet positioned on four corresponding electrodes; subjects held the handles in both hands making contact with corresponding electrodes on the thumbs and palm of each hand. Height, sex and age were entered into the MF-BIA software, and the device collected weight. Subjects stood still for the duration of the assessment.

2.4 | Statistical analysis

Validity statistics were utilized to evaluate the raw differences between MF-BIA and the 4C criterion model for men, women and the total sample. Total body %fat, FM and FFM were evaluated. Constant error (CE) was determined as: $CE = \Sigma(MF\text{-}BIA\text{-}4C)/n$; total error (TE) was determined as $TE = \sqrt{\Sigma(MF\text{-}BIA\text{-}4C)^2/n}$; and standard error of the estimate (SEE) determined as: $SEE = SD(4C)\sqrt{1-r^2}$. The mean difference (MF-BIA average–4C average) and 95% limits of agreement were calculated between the MF-BIA and 4C criterion model for men, women and the total sample; individual variability for total body %fat, FM and FFM between devices was evaluated by Bland–Altman plots. Linear regression from the Bland–Altman plots was utilized to assess proportional bias. Validity analyses and Bland–Altman plots were completed using a customized spreadsheet in Microsoft Excel (Version 16.20, Microsoft Corporation). Paired samples t-tests were utilized to evaluate the difference between MF-BIA and the 4C model criterion. Qualitative evaluation was included based on categorization originally proposed by Lohman et al.: ideal, excellent, very good, good, fairly good, fair and poor (Lohman, 1992). Statistical significance was set at an alpha level of 0.05. Analyses were completed using SPSS statistics software (Version 25, IMB Corporation).

3 | RESULTS

For the total sample, the absolute prediction error was the highest for %fat (CE = 1.6%; TE = 4.2%; SEE = 3.9%) compared to FM (CE = 1.0 kg; TE = 2.4 kg; SEE = 2.2 kg) and FFM (CE = 1.0 kg; TE = 2.4 kg; SEE = 2.2 kg) (Table 2). According to the subjective rating scale by Heyward and Wagner (Heyward et al.,), the TE and SEE validity statistics for %fat suggest fairly good agreement, and ideal to excellent agreement for FM and FFM between the MF-BIA and 4C criterion for the total sample. Bland–Altman plots for total body %fat, FM and FFM separated by total sample and each sex are displayed in Figures 1-3, demonstrating greater bias in individuals

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TABLE 2 Validity statistics for the total sample, men and women for MF-BIA compared to 4C

	Total Sample			Men			Women		
Absolute Error	CE	TE	SEE	CE	TE	SEE	CE	TE	SEE
%fat	1.63	4.16	3.85	1.40	2.54	2.16	1.73	4.73	4.42
FM	0.97	2.38	2.18	0.99	1.86	1.60	0.97	2.59	2.42
FFM	−0.97	2.38	2.18	−0.99	1.86	1.58	−0.97	2.59	2.31

Abbreviations: CE, constant error (kg); FFM; TE, total error (kg); FM, fat-free mass; Per cent body fat, %fat, fat mass; SEE, standard error of the estimate (kg).

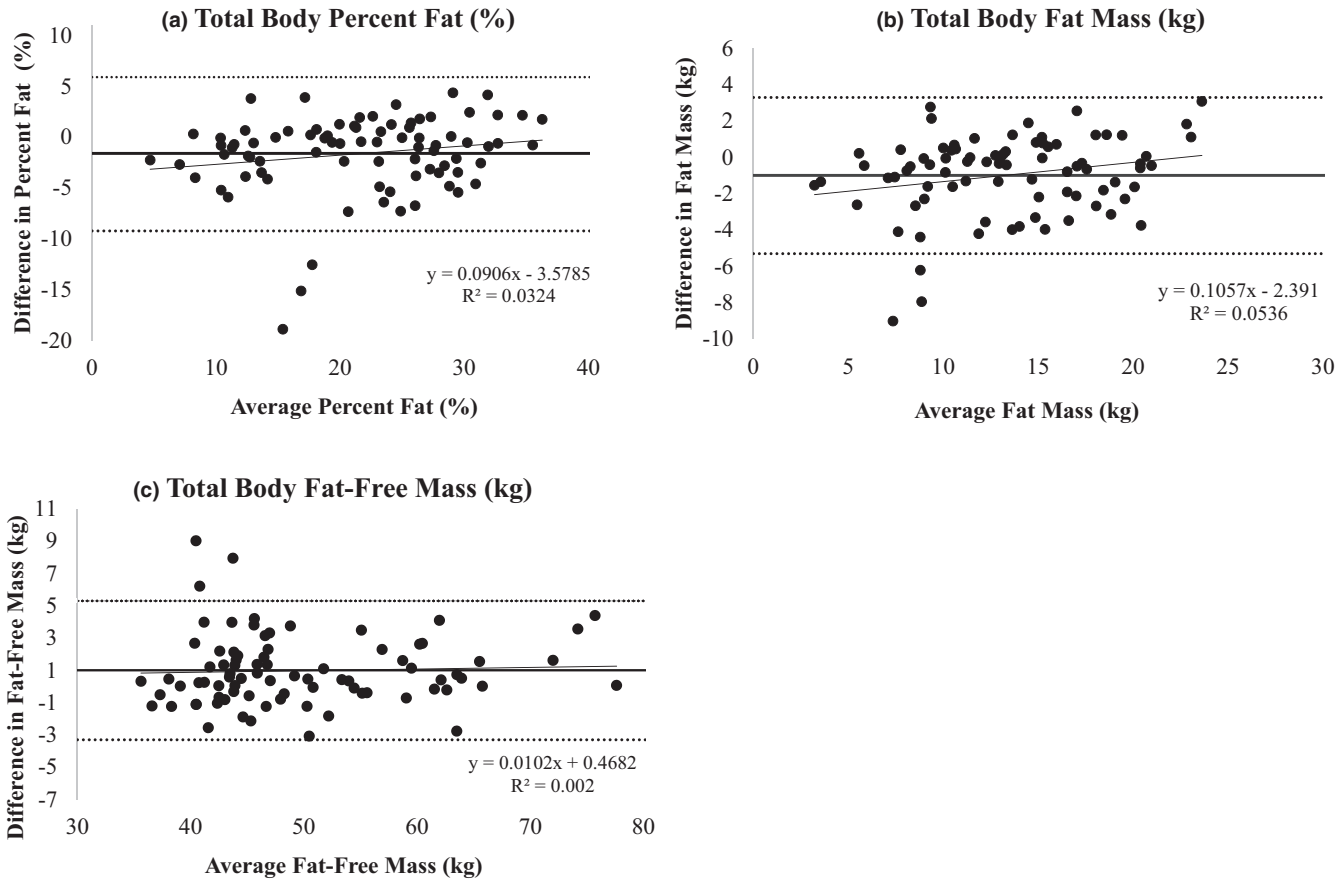

FIGURE 1 Difference between MF-BIA and 4C (MF-BIA–4C) for the entire sample. The constant error (CE) is represented by a solid line; 95% upper and lower limits are represented by dashed lines. The line of regression is represented by a solid line. (a) Per cent body fat. (b) Fat Mass. (c) Fat-Free Mass

with lower levels of %fat, FM and FFM. The proportional bias was significant for total body %fat, FM and FFM within the total sample ($p < 0.001$), men ($p < 0.001$) and women ($p < 0.001$). Significant differences were found between MF-BIA and the 4C criterion model estimates for %fat ($p < 0.001$), FM ($p < 0.001$) and FFM ($p < 0.001$) for the total sample (Table 3).

For the men, the absolute prediction error was the highest for %fat (CE = 1.4%; TE = 2.5%; SEE = 2.2%) compared to FM (CE = 1.0 kg; TE = 1.9 kg; SEE = 1.6 kg) and FFM (CE = 1.0 kg; TE = 1.9 kg; SEE = 1.6 kg). The agreement between the MF-BIA and 4C criterion for %fat ranged from excellent to ideal, and FFM was deemed ideal.

For the women, the absolute prediction error was also the highest for %fat (CE = 1.7%; TE = 4.7%; SEE = 4.4%) compared to FM (CE = 1.0 kg; TE = 2.6 kg; SEE = 2.4 kg) and FFM (CE = 1.0 kg; TE = 2.6; SEE = 2.3 kg). The agreement between the MF-BIA and 4C criterion for %fat ranged from fair to fairly good, and FFM ranged from good to very good.

4 | DISCUSSION

Multi-compartment body composition equations are known to improve the accuracy of body composition estimates (Heymsfield et al.,

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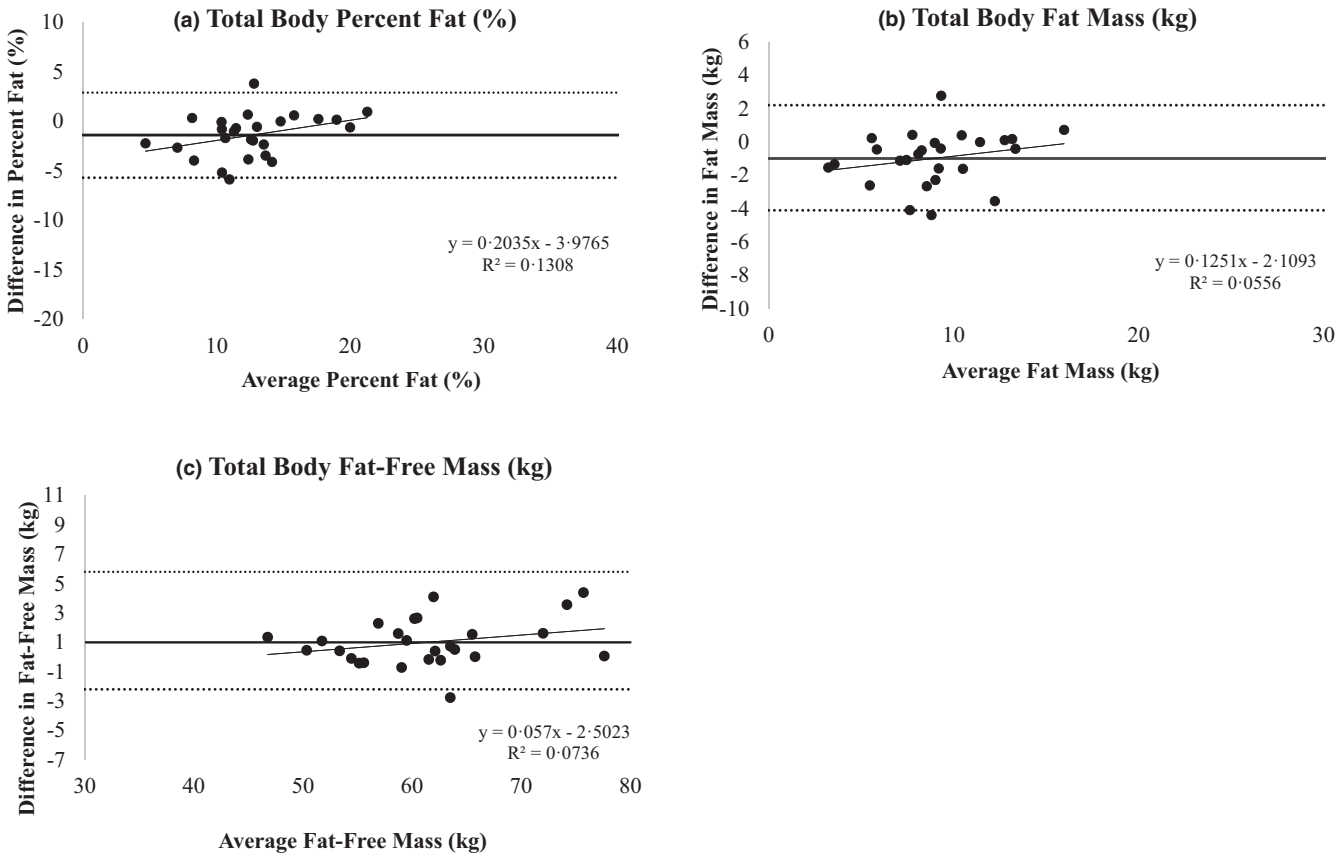

FIGURE 2 Difference between MF-BIA and 4C (MF-BIA–4C) for the men. The constant error (CE) is represented by a solid line; 95% upper and lower limits are represented by dashed lines. The line of regression is represented by a solid line. (a) Per cent body fat. (b) Fat Mass. (c) Fat-Free Mass

1990; Smith-Ryan et al., 2017; Wang et al., 1998). The variety of body composition assessment methods available differ in cost, time to conduct the test, the number of devices needed and accuracy (Heymsfield et al., 1990). When multiple devices are not accessible, 2C model devices are commonly used to estimate body composition (Esco et al., 2015). Results from the present study reveal discrepancies between the MF-BIA and 4C model, which ultimately demonstrate greater agreement in men compared to women. For the total sample, the MF-BIA provided fairly good agreement when estimating %fat (TE = 4.2%; SEE = 3.9%) compared to a 4C model. Validity values were similar for total body FM (TE = 2.4 kg; SEE = 2.2 kg) and FFM (TE = 2.4 kg; SEE = 2.2 kg). The agreement between MF-BIA and the 4C criterion was found to be stronger for men compared to women; within the combined total sample, there was greater agreement for FM and FFM compared to %fat. For the men, the MF-BIA provided excellent agreement for %fat (TE = 2.5%; SEE = 2.2%) and ideal agreement for total body FFM (TE = 1.9 kg; SEE = 1.6 kg), and total body FM was similar (TE = 1.9 kg; SEE = 1.6 kg) (Heyward et al.,). For the women, the MF-BIA provided 'fair' agreement for %fat (TE = 4.7%; SEE = 4.4%) and 'good/very good' agreement for total body FFM (TE = 2.6; SEE = 2.3 kg), with similar values for total body FM (TE = 2.6 kg; SEE = 2.4 kg). Limits of agreement demonstrate the potential inaccuracies in women, when using InBody 770 compared to a 4C model. (Heyward et al.,).

Widely accepted as feasible methods to assess body composition, various BIA devices have been previously validated against DXA technology predominately in athletes (Ling et al., 2011; Nickerson et al., 2019; Raymond et al., 2017; Schoenfeld et al., 2018; Shafer et al., 2009); however, little is known about MF-BIA agreement to higher standard criterion methods specifically in a normal-weight adult population. In healthy adults (average BMI = 25.0 ± 3.6 kg/m^2; average FFM = 55.4 ± 11.8 kg), a similar stand-up MF-BIA (SECA 514/515) was validated against a 4C criterion for FFM estimates and demonstrated similar validity values (TE: 2.0 kg) compared to the present study (Bosy-Westphal et al., 2013). Additional data reported that a similar MF-BIA device produced inaccurate estimates of total body %fat, FM and FFM compared to a 4C model when assessing male and female body builders (average %fat = 17.9 ± 4.7%; FM = 14.5 ± 4.3 kg; FFM = 67.8 ± 15.3 kg) (Graybeal et al., 2020). This previous study reported the largest TE values for %fat estimates when compared to FM and FFM values in women (TE = 4.5%), men (TE = 5.2%) and the total combined sample (TE = 5.0%), which are similar error rates to the present study (Graybeal et al., 2020). For women, the TE reported from this previous study was similar to ours for %fat, FM and FFM; in contrast, our study identified lower TE values for %fat, FM and FFM for normal-weight men, and the total sample, illustrating better agreement in a less muscular sample. Conflicting findings between studies may be due to a difference in

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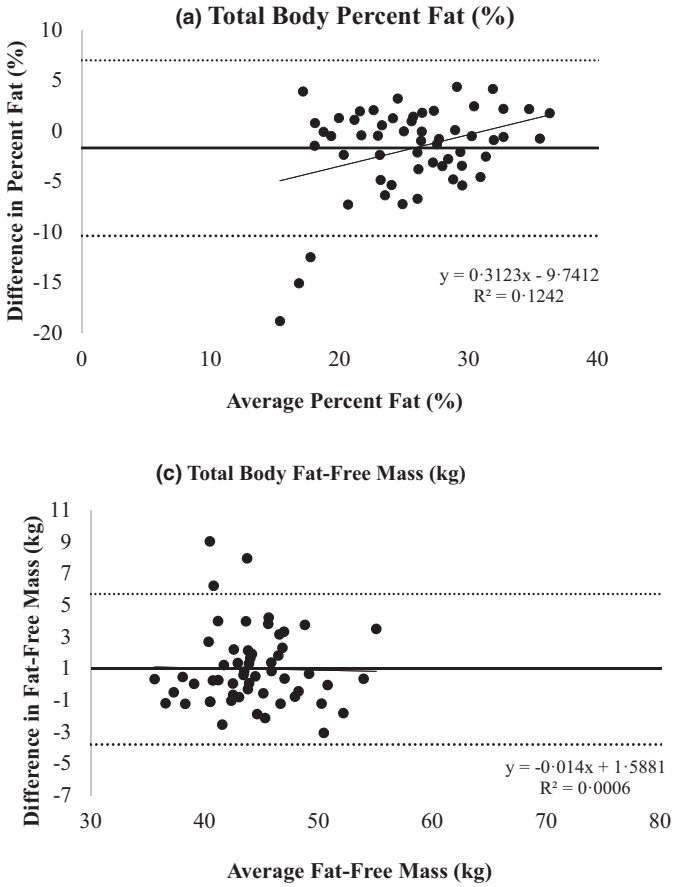

FIGURE 3 Difference between MF-BIA and 4C (MF-BIA–4C) for the women. The constant error (CE) is represented by a solid line; 95% upper and lower limits are represented by dashed lines. The line of regression is represented by a solid line. (a) Per cent body fat. (b) Fat Mass. (c) Fat-Free Mass

TABLE 3 Mean and standard deviation for %fat, FFM and FM for MF-BIA and 4C [mean ± standard deviation (SD)]

MF-BIA	%fat	FM (kg)	FFM (kg)
Total Sample (*n* = 82)	22.3 ± 7.5	13.9 ± 4.7	49.1 ± 9.6
Men (*n* = 26)	13.4 ± 3.6	9.4 ± 2.9	60.7 ± 7.4
Women (*n* = 56)	26.5 ± 4.7	15.9 ± 3.8	43.8 ± 4.3
4C	**%fat**	**FM (kg)**	**FFM (kg)**
Total Sample (*n* = 82)	20.7 ± 8.2	12.9 ± 5.1	50.1 ± 9.7
Men (*n* = 26)	12.0 ± 4.4	8.5 ± 3.3	61.7 ± 7.8
Women (*n* = 56)	24.8 ± 6.2	15.0 ± 4.5	44.7 ± 4.3

Abbreviations: FM, fat mass; FFM, fat-free mass; %fat. Per cent body fat

sample size (*n* = 17 men; *n* = 10 women) compared to the current study (*n* = 26 men; 56 women), and the difference in muscular build between study populations.

Due to physiological differences, current literature in athletic and general population samples implies that the validity of body composition estimates may vary between sex and should therefore be interpreted separately (Brewer et al., 2019; Gibson et al., 2008). Previous validation in men (average %fat = 13.7 ± 4.3%) suggested that a MF-BIA (Xitron 4000B) was not a valid method to evaluate body composition in male athletes reporting significant overestimation of %fat by a BIA device compared to a 4C model (Mean difference [MD]: 12.1 ± 4.0%) in water polo athletes (Andreoli et al., 2004). Although the present study also reported significant overestimation of %fat by a MF-BIA device, our results produced greater agreement between devices (MD: 1.40 ± 2.1%) (Heymsfield & Waki, 1991). Another study utilizing a single-frequency BIA reported that the largest error in %fat occurred when assessing lean men (average %fat = 16.6 ± 3.8%) (Van Marken Lichtenbelt et al., 2004); the current investigation found excellent agreement for %fat estimates between the MF-BIA and 4C model when assessing

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normal-weight men, supporting the benefits of using multiple frequencies in BIA estimation.

Data evaluating the validity of MF-BIA technology to a higher standard criterion in women are limited. The majority of available literature utilizes DXA as the criterion to validate MF-BIA technology (Bolanowski & Nilsson, 2001; De Lorenzo et al., 2000; Esco et al., 2015; Ling et al., 2011; Shafer et al., 2009). One previous study comparing a MF-BIA device to a 4C model in female body builders reported a small overestimation of %fat (MD: 0.9 ± 0.5%), similar to the present study evaluating normal-weight females (MD: 1.7 ± 4.5%) (Graybeal et al., 2020). The same trend was observed with FM from the previous study (MD: 0.6 ± 0.1 kg) (Graybeal et al., 2020) compared to the greater variation in the present study in normal-weight females (MD: 2.4 ± 1.0 kg) The discrepancy between studies may result from the body builders possessing greater amounts of FFM and lower FM, compared to the normal-weight females in the present study. It should be noted that TE values were nearly identical between studies for MF-BIA analyses. There appeared to be good agreement between methods for FFM in the previous sample (MD: 0.60 ± 0.5 kg); estimates from the current study were similar, but more variable (MD: 0.97 ± 2.4 kg) (Graybeal et al., 2020). In elderly women, BIA technology underestimated FM (MD: 4.5 ± 2.4 kg) and %fat (MD: 7.6 ± 2.9%) when compared to DXA, which is contrary to the present study (Bolanowski & Nilsson, 2001). Data in obese women suggest that BIA technology may also not be an accurate method, compared to DXA, to obtain body composition estimates (Panotopoulos et al., 2001). Validity differences between men and women may be attributed to sex-specific fat distribution patterns, as well as differences in muscle quality (Shafer et al., 2009).

It is important to highlight key factors that may influence the accuracy of body composition estimate, particularly with MF-BIA technology, to better identify how the accuracy of devices may be improved. The hydration status of a participant will impact the validity of these devices; if a subject is dehydrated or hyperhydrated, violation of standard assumptions may occur (Schoenfeld et al., 2018; Smith-Ryan et al., 2017; Wang et al., 1999; Wilson et al., 2013). Hydration status was not directly measured in the present study which is a limitation; however, the effect of hydration status would carry-over to the 4C model, likely resulting in similar implications. In addition to hydration, acute food and fluid intake (Tinsley et al., 2017) and exercise (Nana et al., 2012; Nana et al., 2013) may significantly alter body composition measurements, particularly for BIA (Nickerson et al., 2019; Schoenfeld et al., 2018; Tinsley et al., 2017). In attempt to control for such variability, subjects were requested to arrive to the laboratory fasted and euhydrated; however, subject adherence to pre-assessment guidelines was self-reported. Additionally, menstrual cycle was not controlled for, and women were not excluded for taking hormonal contraceptives. The influence of hormonal contraceptives and stage of the menstrual cycle during which that testing occurs may influence TBW, FM, %fat and FFM estimates (Bunt et al., 1989; Lukaski, 2013); however, with this one-time cross-sectional analysis, it is likely to have a negligible effect. While the use of deuterium dilution is the gold standard method for total body water assessment, collection of TBW from the BIS utilized in the present study has been validated and commonly implemented in various research and clinical settings (Buendia et al., 2015; Moon et al., 2008).

5 | PERSPECTIVES

Findings from this study suggest that in healthy adults, the MF-BIA utilized in this study (InBody 770) may produce fairly good estimates of body composition (FM, and FFM) compared to a 4C model; %fat estimates resulted in greater error, yet still acceptable. MF-BIA resulted in greater error for estimates in normal-weight women, particularly when evaluating %fat, resulting in nearly twice as much error. For men, the MF-BIA resulted in 'excellent' agreement but may overestimate %fat by an average of 2.2% and underestimate FFM by an average of 1.6 kg. In women, the MF-BIA may overestimate %fat by an average of 4% and underestimate FFM by an average of 2.3 kg. The greater limits of agreement observed in men may suggest greater bias in leaner individuals when using MF-BIA technology. Practically, MF-BIA device may be a more cost-effective and feasible body composition assessment method, compared to a 4C model; in a mixed sample of men and women, accuracy was 'fairly good' and acceptable. Estimates in women resulted in greater error, particularly for %fat, but may still be acceptable compared to other 2C devices.

CONFLICT OF INTEREST

All authors confirm an absence of conflict of interest.

DATA AVAILABILITY STATEMENT

Deidentified individual data that supports the results will be shared beginning nine to 36 months following publication provided the investigator who proposes to use the data has approval from an Institutional Review Board (IRB), Independent Ethics Committee (IEC), or Research Ethics Board (REB), as applicable, and executes a data use/sharing agreement with the University of North Carolina.

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REFERENCES

Alves, F.D., Souza, G.C., Biolo, A. & Clausell, N. (2014) Comparison of two bioelectrical impedance devices and dual-energy X-ray absorptiometry to evaluate body composition in heart failure. *Journal of Human Nutrition & Dietetics*, *80*(6), 2085–2096.

Andreoli, A., Melchiorri, G., Volpe, S.L., Sardella, F., Iacopino, L. & De Lorenzo, A. (2004) Multicompartment model to assess body composition in professional water polo players. *Journal of Sports Medicine and Physical Fitness*, *44*(1), 38–43.

Bolanowski, M. & Nilsson, B.E. (2001) Assessment of human body composition using dual-energy x-ray absorptiometry and bioelectrical impedance analysis. *Medical Science Monitor*, *7*, 1029–1033.

Bosy-Westphal, A., Schautz, B., Later, W., Kehayias, J.J., Gallagher, D. & Müller, M.J. (2013) What makes a BIA equation unique? Validity of

1475097x, 2021, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/cpf.12700 by United Arab Emirates, Wiley Online Library on [30/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

eight-electrode multifrequency BIA to estimate body composition in a healthy adult population. *European Journal of Clinical Nutrition*, *67*(1), S14–21.

Brewer, G.J., Blue, M.N.M., Hirsch, K.R., Peterjohn, A.M. & Smith-Ryan, A.E. (2019) Appendicular body composition analysis: Validity of bioelectrical impedance analysis compared with dual-energy X-Ray absorptiometry in division I college athletes. *Journal of Strength and Conditioning Research*, *33*(11), 2920–2925.

Buendia, R., Seoane, F., Lindecrantz, K., Bosaeus, I., Gil-Pita, R., Johannsson, G. et al. (2015) Estimation of body fluids with bioimpedance spectroscopy: State of the art methods and proposal of novel methods. *Physiological Measurement*, *36*(10), 2171–2187.

Bunt, J.C., Lohman, T.G. & Boileau, R.A. (1989) Impact of total body water fluctuations on estimation of body fat from body density. *Medicine and Science in Sports and Exercise*, *21*(1), 96–100.

Collins, A.L. & McCarthy, H.D. (2003) Evaluation of factors determining the precision of body composition measurements by air displacement plethysmography. *European Journal of Clinical Nutrition*, *57*(6), 770–776.

De Lorenzo, A., Bertini, I., Iacopino, L., Pagliato, E., Testolin, C. & Testolin, G. (2000) Body composition measurement in highly trained male athletes: A comparison of three methods. *Journal of Sports Medicine and Physical Fitness*, *40*(2), 178.

Demerath, E.W., Guo, S.S., Chumlea, W.C., Towne, B., Roche, A.F. & Siervogel, R.M. (2002) Comparison of percent body fat estimates using air displacement plethysmography and hydrodensitometry in adults and children. *International Journal of Obesity*, *26*(3), 389–397.

Esco, M.R., Snarr, R.L., Leatherwood, M.D., Chamberlain, N.A., Redding, M.L., Flatt, A.A. et al. (2015) Comparison of total and segmental body composition using DXA and multifrequency bioimpedance in collegiate female athletes. *Journal of Strength and Conditioning Research*, *29*(4), 918–925.

Forslund, A.H. (1998) Evaluation of modified multicompartment models to calculate body composition in healthy males. *Applied Radiation and Isotopes*, *49*(5-6), 599–601.

Gibson, A.L., Holmes, J.C., Desautels, R.L., Edmonds, L.B. & Nuudi, L. (2008) Ability of new octapolar bioimpedance spectroscopy analyzers to predict 4-component-model percentage body fat in Hispanic, black, and white adults. *American Journal of Clinical Nutrition*, *87*(2), 332–338.

Graybeal, A.J., Moore, M.L., Cruz, M.R. & Tinsley, G.M. (2020) Body composition assessment in male and female bodybuilders: A 4-compartment model comparison of dual energy x-ray absorptiometry and impedance based devices. *Journal of strength and conditioning Research*, *34*(6), 1676–1689.

Heymsfield, S.B., Lichtman, S., Baumgartner, R.N., Wang, J., Kamen, Y., Aliprantis, A. et al. (1990) Body composition of humans: Comparison of two improved four-compartment models that differ in expense, technical complexity, and radiation exposure. *American Journal of Clinical Nutrition*, *52*(1), 52–58.

Heymsfield, S.B. & Waki, M. (1991) Body composition in humans: Advances in the development of multicompartment chemical models. *Nutrition Reviews*, *49*(4), 97–108.

Heyward, V.H. & Wagner, D.R. (2004) *Applied body composition assessment*. In: Heyward, V.H.W.D. (eds). (2nd ed). Champaign: Human Kinetics Publishers, Inc; 19–21.

Kyle, U.G., Bosaeus, I., De Lorenzo, A.D., Deurenberg, P., Elia, M., Gómez, J.M. et al. (2004) Bioelectrical impedance analysis–Part I: Review of principles and methods. *Clinical Nutrition*, *23*(5), 1226–1243.

Ling, C.H.Y., De Craen, A.J.M., Slagboom, P.E., Gunn, D.A., Stokkel, M.P.M., Westendorp, R.G.J. et al. (2011) Accuracy of direct segmental multi-frequency bioimpedance analysis in the assessment of total body and segmental body composition in middle-aged adult population. *Clinical Nutrition*, *30*(5), 610–615.

Lohman, T.G. (1992) *Current issues in exercise science series: Advances in body composition assessment*. Champaign: Human Kinetics Publishers, Inc, 3–4.

Lukaski, H.C. (1997) Validation of body composition assessment techniques in the dialysis population. *ASAIO Journal*, *43*, 251–255.

Lukaski, H.C. (2013) Evolution of bioimpedance: A circuitous journey from estimation of physiological function to assessment of body composition and a return to clinical research. *European Journal of Clinical Nutrition*, *67*(1), S2–S9.

Modlesky, C.M., Cureton, K.J., Lewis, R.D., Prior, B.M., Sloniger, M.A. & Rowe, D.A. (1996) Density of the fat-free mass and estimates of body composition in male weight trainers. *Journal of Applied Physiology*, *80*(6), 2085–2096.

Moon, J.R., Tobkin, S.E., Roberts, M.D., Dalbo, V.J., Kerksick, C.M., Bemben, M.G. et al. (2008) Total body water estimations in healthy men and women using bioimpedance spectroscopy: A deuterium oxide comparison. *Nutrition & Metabolism*, *5*(1), 7.

Nana, A., Slater, G.J., Hopkins, W.G. & Burke, L.M. (2012) Effects of daily activities on dual-energy X-ray absorptiometry measurements of body composition in active people. *Medicine and Science in Sports and Exercise*, *44*(1), 180–189.

Nana, A., Slater, G.J., Hopkins, W.G. & Burke, L.M. (2013) Effects of exercise sessions on DXA measurements of body composition in active people. *Medicine and Science in Sports and Exercise*, *45*(1), 178–185.

Nickerson, B.S., Tinsley, G.M. & Esco, M.R. (2019) Validity of field and laboratory three-compartment models in healthy adults. *Medicine and Science in Sports and Exercise*, *51*(5), 1032–1039.

Panotopoulos, G., Ruiz, J.C., Guy-Grand, B. & Basdevant, A. (2001) Dual x-ray absorptiometry, bioelectrical impedance, and near infrared interactance in obese women. *Medicine and Science in Sports and Exercise*, *33*(4), 665–670.

Raymond, C.J., Dengel, D.R. & Bosch, T.A. (2017) Total and segmental body composition examination in collegiate football players using multifrequency Bia and Dxa. *Journal of Strength and Conditioning Research*, *32*(3), 772–782.

Ritz, P., Vol, S., Berrut, G., Tack, I., Arnaud, M.J. & Tichet, J. (2008) Influence of gender and body composition on hydration and body water spaces. *Clinical Nutrition*, *27*(5), 740–746.

Schoenfeld, B.J., Nickerson, B.S., Wilborn, C.D., Urbina, S.L., Hayward, S.B., Krieger, J. et al. (2018) Comparison of multifrequency bioelectrical impedance vs. dual-energy X-ray absorptiometry for assessing body composition changes after participation in a 10-week resistance training program. *Journal of Strength and Conditioning Research*, *34*(3), 678–688.

Shafer, K.J., Siders, W.A., Johnson, L.A.K. & Lukaski, H.C. (2009) Validity of segmental multiple-frequency bioelectrical impedance analysis to estimate body composition of adults across a range of body mass indexes. *Nutrition*, *25*(1), 25–32.

Smith-Ryan, A.E., Mock, M.G., Ryan, E.D., Gerstner, G.R., Trexler, E.T. & Hirsch, K.R. (2017) Validity and reliability of a 4-compartment body composition model using dual energy x-ray absorptiometry-derived body volume. *Clinical Nutrition*, *36*(3), 825–830.

Tinsley, G.M., Moore, M.L., Silva, A.M. & Sardinha, L.B. (2020) Cross-sectional and longitudinal agreement between two multifrequency bioimpedance devices for resistance, reactance, and phase angle values. *European Journal of Clinical Nutrition*, *74*(6), 900–911.

Tinsley, G.M., Morales, E., Forsse, J.S. & Grandjean, P.W. (2017) Impact of acute dietary manipulations on DXA and BIA body composition estimates. *Medicine and Science in Sports and Exercise*, *49*(4), 823–832.

Van Marken Lichtenbelt, W.D., Hartgens, F., Vollaard, N.B.J., Ebbing, S. & Kuipers, H. (2004) Body composition changes in bodybuilders: A method comparison. *Medicine and Science in Sports and Exercise*, *36*(3), 490–497.

1475097x, 2021, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/cpf.12700 by United Arab Emirates, Wiley Online Library on [30/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Wang, Z.M., Deurenberg, P., Guo, S.S., Pietrobelli, A., Wang, J., Pierson, R.N. et al. (1998) Six-compartment body composition model: Inter-method comparisons of total body fat measurement. *International Journal of Obesity*, *22*(4), 329–337.

Wang, Z.M., Deurenberg, P., Wang, W., Pietrobelli, A., Baumgartner, R.N. & Heymsfield, S.B. (1999) Hydration of fat-free body mass: Review and critique of a classic body- composition constant. *American Journal of Clinical Nutrition*, *69*(5), 883–841.

Wang, Z.M., Xavier Pi-Sunyer, F., Kotler, D.P., Wielopolski, L., Withers, R.T., Pierson, R.N. et al. (2002) Multicomponent methods: Evaluation of new and traditional soft tissue mineral models by in vivo neutron activation analysis. *American Journal of Clinical Nutrition*, *76*(5), 968–974.

Wilson, J.P., Fan, B. & Shepherd, J.A. (2013) Total and regional body volumes derived from dual-energy x-ray absorptiometry output. *Journal of Clinical Densitometry*, *16*(3), 368–373.

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